AEGIS-RF: Confidence-Aware, Resource-Adaptive RF Spectrum Intelligence

On Microchip PolarFire® SoC Icicle Kit (MPFS250T)

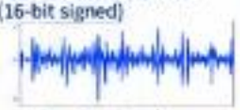
Continuous Full-Band Candidate Awareness
Fixed 2048-bin scan active in all modes; only downstream candidate analysis, feature extraction and reporting are adapted.

1 PolarFire FPGA Fabric – Deterministic Real-Time Fast Loop (Continuous Full-Band Candidate Scan)

Complex I/Q Input Source (From PC via Ethernet / Recorded Files)



MATLAB-Generated / Recorded Complex I/Q (16-bit signed)



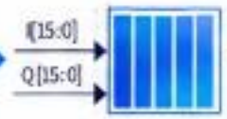
- Sample format : I[15:0], Q[15:0]
- Sample order : I, Q interleaved
- Frame length : 2048 samples
- Interface : AXI4-Stream (TLAST per frame)
- Valid/Ready handshake

Optional Future Extension (RF Front-End / SDR)



SDR (e.g., PlutoSDR / USRP) Real RF input (optional)

Input FIFO & Frame Control (AXI4-Stream)



- AXI4-Stream slave
- Frame length: 2048
- tvalid, tready, tlast, tuser
- Back-pressure handling
- 16-bit complex samples

Fixed 2048-Point FFT (Microchip CoreFFT v7.1)



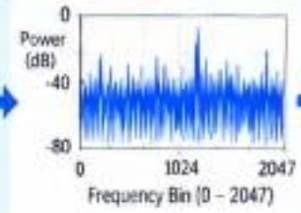
- Point size : 2048
- Input : 16-bit complex
- Output : 32-bit complex (scaled)
- Streaming architecture
- Frame sync : tlast from FFT
- Scaling and overflow control (configurable as per user guide)

Magnitude Squaring (Power Spectrum)

$$|X[k]|^2 = Re^2 + Im^2$$

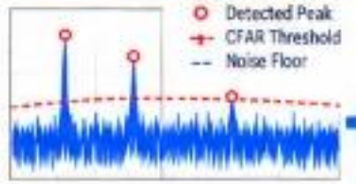
- 32-bit magnitude output
- Configurable scaling
- Saturation handling

Continuous 2048-Bin Full-Band Candidate Scan



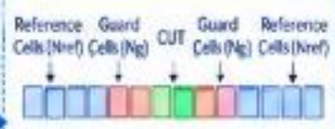
- Scan all 2048 bins (always active)
- Generate candidate list
- Peaks above noise floor

Sub-Band CA-CFAR Detection



- Cell-Averaging CFAR (CA-CFAR)
- Configurable reference cells (Nref)
- Configurable guard cells (Ng)
- Threshold = α · noise estimate
- Sub-band windowed processing
- Configurable parameters (via AXI-Lite)

CA-CFAR Window (per CUT)



Noise estimate:

$$\hat{P}_n = \frac{1}{N_{ref}} = \sum_{ref} P_{ref}$$

Threshold: $T = \alpha \cdot \hat{P}_n$

- Nref, Ng, α: configurable (AXI-Lite)
- Linear or dB domain (configurable)

Statistical Feature Extraction (for Selected Candidates)

- Envelope variance
- Average power (PAPR)
- Spectral centroid
- Bandwidth (occupied BW)
- Spectral flatness
- Temporal features (persistence, stability)
- Additional configurable features

2 Interconnect & Memory Infrastructure (Based on PolarFire SoC LCArchitecture)

2 GB LPDDR4 Memory (On-Board, Shared System Memory)



Icicle Kit

2 GB LPDDR4 (Off-Chip, Shared Memory)

- Accessible by RISC-V via system interconnect
- Accessible by FPGA via AXI
- Cache coherent access (via coherent interconnect)
- Address map as per user guide

Ping-Pong Buffer Structure (Feature Descriptors in DDR)

Buffer A (Active – FPGA Writing)	Alternate Each Frame (Ping-Pong)	Buffer B (Standby / CPU Reading)
• Candidate list • Feature vectors • Timestamps • Metadata	⟷	• Candidate list • Feature vectors • Timestamps • Metadata

Buffer Ownership State Machine

FREE (Available) → FPGA WRITING (DMA in progress) → COMPLETE (IRQ to CPU) → CPU READING (Process data) → FREE (Available)

System Interconnect (PolarFire SoC)

- AXI Interconnect (Amba AXI4)
- AXI4 (Data Path) (FPGA ↔ LPDDR4 via DMA)
- AXI4-Lite (Control/Status)
- Interrupt Controller (Fabric → RISC-V)
- Cache Coherency (as per user guide)

DMA / Shared-Memory Transfer Path (e.g., CoreAXI4DMA)

- 64-bit AXI4 master (per user guide)
- Scatter-gather DMA support
- Descriptor-based transfers
- Memory-mapped buffers in LPDDR4
- Cache coherency via MSS interconnect
- Interrupt on frame/buffer complete
- Error status (overflow, timeout)

IRQ Interrupt Line (Buffer Complete / Error)

RISC-V Coreplex (Quad-core SiFive U54)

Core 0 | Core 1 | Core 2 | Core 3

- 64-bit RISC-V U54 cores
- Linux (as per user guide)
- SMP support

AXI-Lite Bus (Example Base Address: 0x4000_0000 – verify in design)

Shadow Registers (Software Writable)

- CFAR parameters
- Window size (Nref, Ng)
- Feature selection
- Mode parameters

sof_sync (Start-of-Frame) — Latch at frame boundary

Active Registers (Hardware Used)

- Latched on sof_sync
- Used by FPGA fabric in real-time
- Safe update mechanism

3 Hardened RISC-V Processing Subsystem – Supervisory Slow Loop (Linux on Quad-core 64-bit U54)

Linux User Space (Spectrum Intelligence Application)



- Mode management (Coarse / Standard / Detailed)
- Parameter adaptation
- Telemetry and logging
- Spectrum visualitation
- Performance monitoring
- Error handling and safe fallback

Driver-Managed DMA Buffers (Memory Mapping with Cache Coherency)



- Linux kernel driver (reference design)
- mmap() to DMA buffers
- Cache coherence (as per user guide)
- Descriptor status and interrupts
- Buffer management (ping-pong)
- Optional: DMAEngine / UIO driver (depending on implementation)

Confidence Metric Engine (Feature Normalization and Weighted Fusion)

$$C_{det} = w_1 \cdot PNR + w_2 \cdot Persistence + w_3 \cdot Compactness$$

- PNR : Peak-to-noise ratio (normalized [0, 1])
- Persistence : Temporal consistency (normalized)
- Compactness : Spectral compactness (normalized)
- Weights w_1, w_2, w_3 : configurable
- Thresholds : calibrated using labelled test data

Spectrum Activity FSM (Adaptive Processing Control)

Coarse (Low Activity) ⟷ Detailed (Weak/Unstable)

Escalate if $C_{det} \geq T_{high}$ (configurable)

De-escalate if $C_{det} < T_{low}$

AND dwell time $\geq N_f$ frames (configurable)

Fault Handling & Safe Fallback



- DMA error / buffer overflow
- Invalid parameters
- Dropped frames
- Unresponsive controller
- Status flags and logging
- Switch to safe mode (low-complexity scan)
- Keep full-band scan active where possible

4 Peripherals & Telemetry (On-Board Resources and External Monitoring)

I2C Power Monitoring (On Board – per user guide)



Microchip PAC1934 4-Channel DC Rail Monitor

- Measure board rails (verify rail names)
- VDD1 (e.g., 1.0 V, board dependent)
- VDD2 (e.g., 1.2 V or 2.5 V, board dependent)
- VDD3 (e.g., 2.5 V, board dependent)
- VDD4 (e.g., 3.3 V, board dependent)
- Current and power telemetry via I2C

I2C

RISC-V Linux (On Icicle Kit)



- Read power/voltage/current (PAC1934)
- Log telemetry data
- System monitoring and logging
- No thermal telemetry unless Provide metrics to dashboard

Dual Gigabit Ethernet (On Board)



- 2 × 1 Gbps Ethernet (RJ45)
- Network to host PC
- Real-time data streaming
- Visualte configuration

TCP/IP



Remote PC Monitoring Dashboard (Live Spectrum and Telemetry)

- Real-time spectrum display
- Detection events and confidence
- Performance metrics
- Power telemetry (PAC1934)
- Mode status and logs
- Remote configuration (optional)